Week 9 Lab 3: Nozzle Data Alignment Audit

FlowMLLab – AI in Fluid Mechanics

Purpose and scope

Extend the Week 4 branch–trunk data contract to moving-throat nozzle grids. This CPU exercise uses an ideal-gas, inviscid, isentropic, choked, shock-free quasi-one-dimensional reference. It does not train a neural model or validate multidimensional CFD.

Three objects, one physical pairing

For nozzle c and query j,

$$B_c = [A_c(\xi_1)/A_t, \ldots, A_c(\xi_{41})/A_t, p_{0,c}, T_{0,c}], \qquad (B_c, x_{cj}) \mapsto q_c(x_{cj}).$$

Here $A_c(\xi)$ is the cross-sectional area of nozzle c at axial position ξ, A_t the throat area (so A_c/A_t is the area ratio that the isentropic relation uses), $p_{0,c}$ and $T_{0,c}$ the stagnation pressure and temperature of the inflow, x_{cj} the j-th axial query point of nozzle c, and x_t the throat position, which moves between nozzles. The 41 sensor locations ξ_i are the same for every nozzle: this fixed sampling is what lets one branch input vector describe every geometry in the family (the number 41 is a choice; what matters is that it never changes). The geometry values vary. Query grids may differ; here each includes its own throat exactly. Targets are $q = (M, p, T, \rho, u)$: Mach number, pressure, temperature, density and velocity. The branch and target case IDs must agree, and the query must equal the source target coordinate in the same units.

Intentional failures

The notebook first repeats the first nozzle's query grid for every nozzle. All array shapes remain correct, but the coordinate audit rejects the pairing. It then incorrectly uses the case-dependent query grids as fixed branch sensors; the sensor audit rejects this too. These are expected caught errors, not failed notebook execution.

Repair and reference checks

Keep each target's own query location; evaluate branch geometry at the common 41 sensors. Alternatively, evaluate all reference fields directly at common queries. If CFD fields are interpolated instead, measure interpolation error separately. Check positivity, $|M(x_t) - 1|$, area–Mach consistency and

$$C_{\dot{m}} = \frac{\mathrm{std}(\rho u A/A_t)}{\mathrm{mean}(\rho u A/A_t)}.$$

These checks verify the reference. Passing them does not demonstrate neural accuracy. An audit also cannot discover metadata that was already mislabeled by the exporter; retain independent source coordinates.

From debugging to a model comparison

Split complete physical cases before expanding spatial rows. Fit normalization and any POD basis on selection-training cases only. Validation supports model selection and early stopping. Freeze MLP and DeepONet before a fresh final test; previously inspected cases are diagnostic data. Report per-case relative L_2 mean, standard deviation and maximum; physical violations; and warmed-up full-profile prediction time. A new random seed is not proof of disjoint cases. The two-parameter geometry family is not evidence of arbitrary-shape generalization.

Questions

Why can loss decrease on incorrectly paired data? Why does repeating a branch row differ from tiling the branch matrix? Why need sensors and queries not coincide? What evidence distinguishes metadata consistency, reference validity, and neural accuracy? Explain before adding a more complex architecture.

Week 9 | Reading the evidence

Local accuracy, geometry and the evaluation contract

In the zonal loss

$$\mathcal{L} = \alpha \frac{1}{N_v} \sum_{i \in V} e_i^2 + (1-\alpha) \frac{1}{N_m} \sum_{i \in M} e_i^2,$$

V denotes reverse-flow samples, M the remaining samples, and N_v, N_m their counts. The scalar α weights regional mean errors, rather than the number of pixels in each region. Select it on validation cases, then freeze it before opening final cases. Always show unweighted global AND reverse-flow validation errors beside the weighted fits. A validation guardrail, such as a two-percentage-point global penalty, is a selection rule, not a guarantee on new geometries. Map the largest local error; it may sit at the step lip rather than inside the reverse-flow zone.

The classroom coordinate model is an MLP. A strict branch/trunk model that uses only $(x/H, y/H)$ in its trunk has a different geometry representation and training problem. A worse result is not by itself proof of an architectural limitation. Compare seeds, budget, training error and explicit wall-relative features before attributing the difference to geometry. Do not mix numbers from different runs.

In nozzle registration, a station returned only at grid nodes cannot distinguish sub-cell shock motion. Equal detected stations do not prove identical physical shocks. Report grid spacing, ties and every non-monotone pair. Target-assisted alignment is a representation audit; deployment must use a training-only locator. Historical holdouts are regression cases once inspected, not new blind cases.

Branch sensors and query coordinates have different roles. Fixed branch sensors describe the input consistently across cases. Each query must still be paired with its own target value; equal array shapes alone do not establish that pairing.

Check your understanding. Identify the input information, the reference, the metric denominator and the evaluation split in one worked notebook comparison. Predict which conclusion would change if any one of those four changed, then check your prediction.